



Lesson 2: Data Exploration and Preparation

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

Before we start

Exercise 1 was set last week. We discuss it now.

- The wine-quality workflow, end to end
- What counts is **methodology**, not accuracy
- A show of hands on how you framed it

What today builds on

- Lesson 1's rule: **nothing is learned before the split**
- Lesson 1's tool: `Pipeline`, so the rule is structural, not just discipline
- Today: apply both to data that is actually messy

Today

- Exploratory analysis
- Missing values
- Outliers
- Scaling and encoding
- Feature engineering and pipelines
- Leakage, in a form nothing warns you about

One dataset, all lesson long

2000 synthetic telecom customers. Will they **churn?**

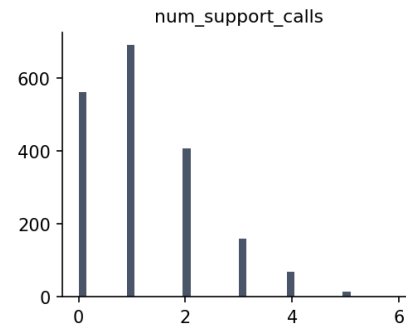
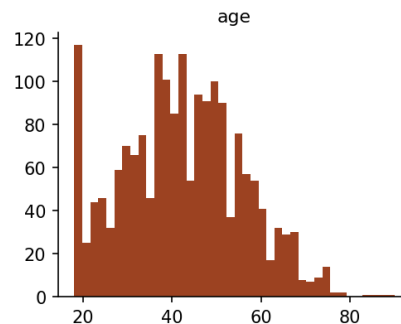
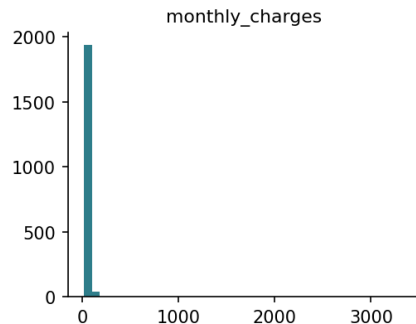
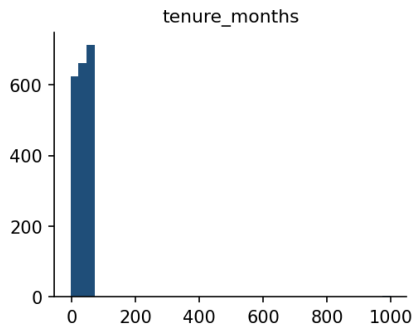
- Numeric: tenure, monthly charges, age, support calls
- Categorical: contract type, region, zip code (493 levels)
- Missing values, outliers, and one column built to carry **no signal**

Look before you touch anything

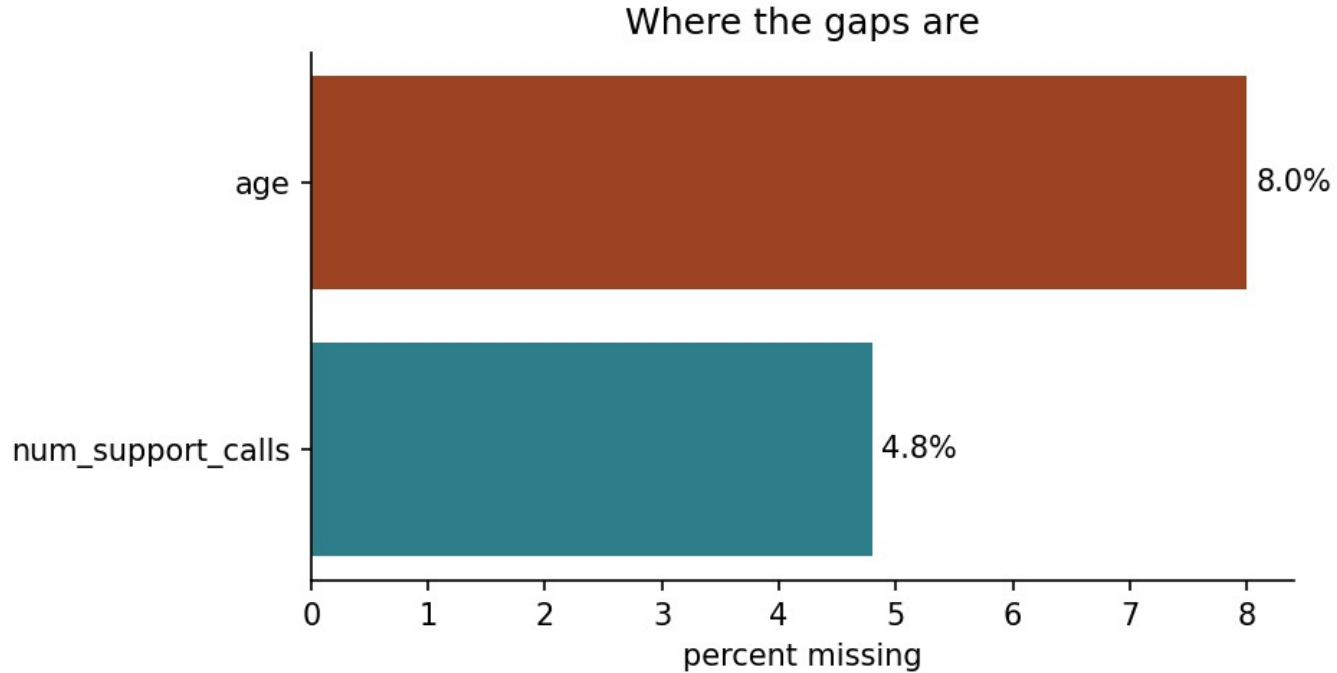
- `.info()`, `.describe()`, class balance, first: always.
- 2000 rows, 8 columns, mixed types
- Churn rate: **19.4%** → baseline accuracy 80.6%
- `tenure_months` max: 999. Not a typo in the slide.

Two of these four have no shape left to read

Raw distributions, before any cleaning



Where the gaps are



Three reasons a value is missing

Three reasons a value can be missing

MCAR

missing completely at random

P(missing)



depends on nothing

MAR

missing at random

P(missing)



depends on an
OBSERVED column

MNAR

missing not at random

P(missing)



depends on the
MISSING value itself

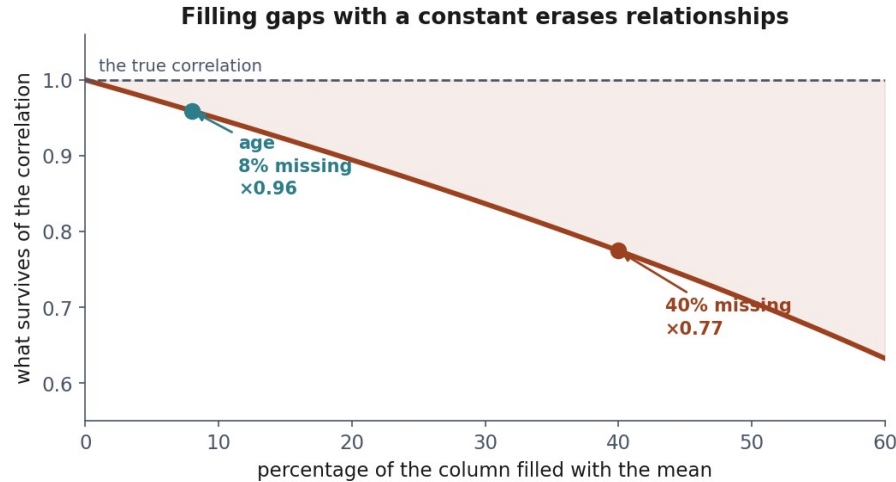
age: missing completely at random

MCAR, for short: some customers just left it blank.
Nothing systematic.

- Safe to impute with a fixed statistic
- The bias question is not “is this fair”: it is “what does filling it cost”

What mean imputation actually costs

Every correlation involving the imputed column shrinks, even under MCAR.



Support calls: missing at random

MAR, for short: missing more often for long-tenure customers - a column we can see explains it.

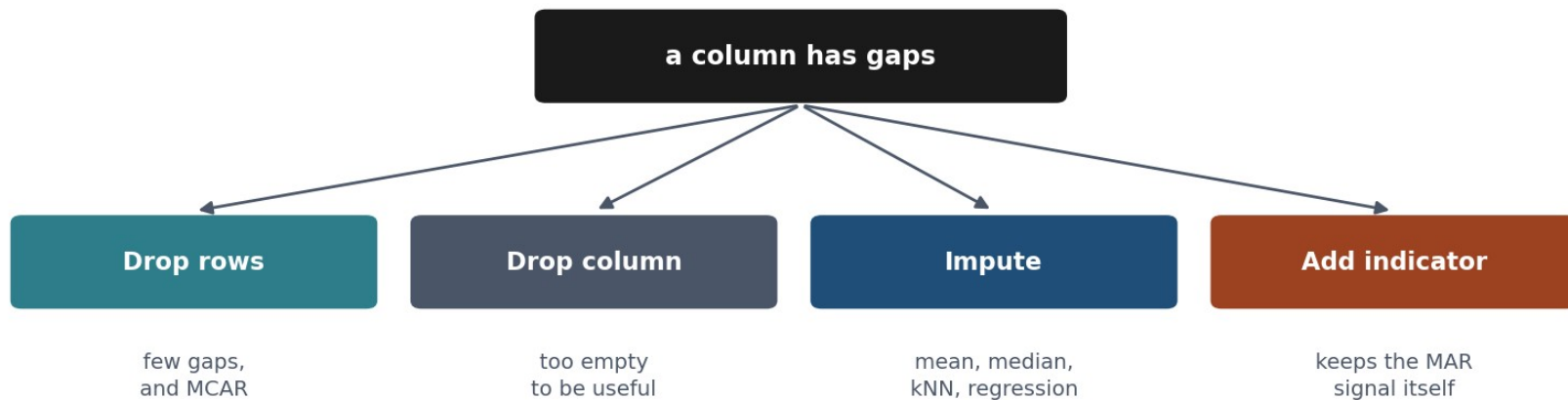
- Treating it as MCAR reweights the sample the wrong way
- A **missingness indicator** column preserves the signal even after filling

What to do about a gap

- **Drop rows:** only if few, and only if MCAR
- **Drop the column:** if it is missing too often to help
- **Impute:** mean/median or most-frequent with `SimpleImputer`, neighbours with `KNNImputer`
- **Add a missingness indicator:** keeps MAR signal alive

Choosing among the four

Which one you pick is a question about why the data is missing



the mechanism decides, and every option except dropping the column learns a statistic, so it belongs inside the training fold

An outlier is far from the rest - not necessarily wrong

- `tenure_months` runs **-3 to 999**; ordinary customers sit at 0 to 72
- Three reasons: a **recording error**, a **rare but genuine** value, a **different population**
- The arithmetic cannot tell which: 3,344.7 may be a typo or a corporate account
- Two rules follow - standard deviations, then quartiles - and both only **flag**

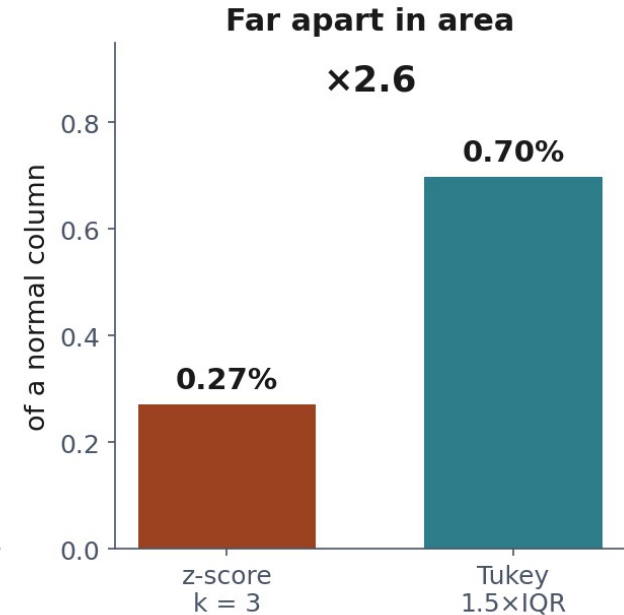
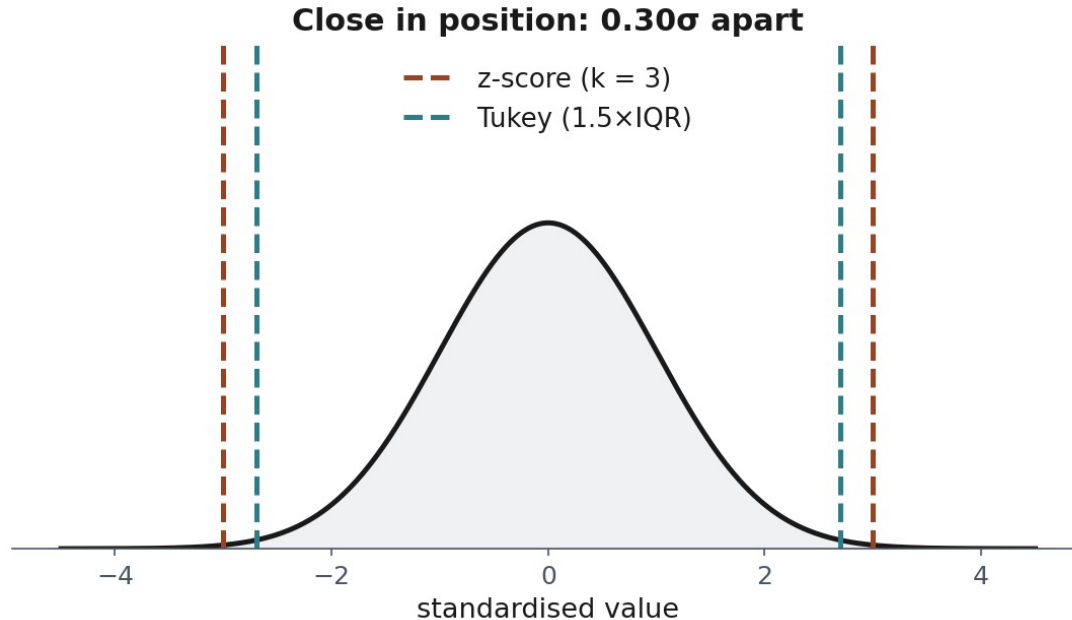
Rule 1: distance in standard deviations, flag beyond $k = 3$

$$Z_i = \frac{x_i - \bar{x}}{s}$$

Rule 2: distance measured in quartiles

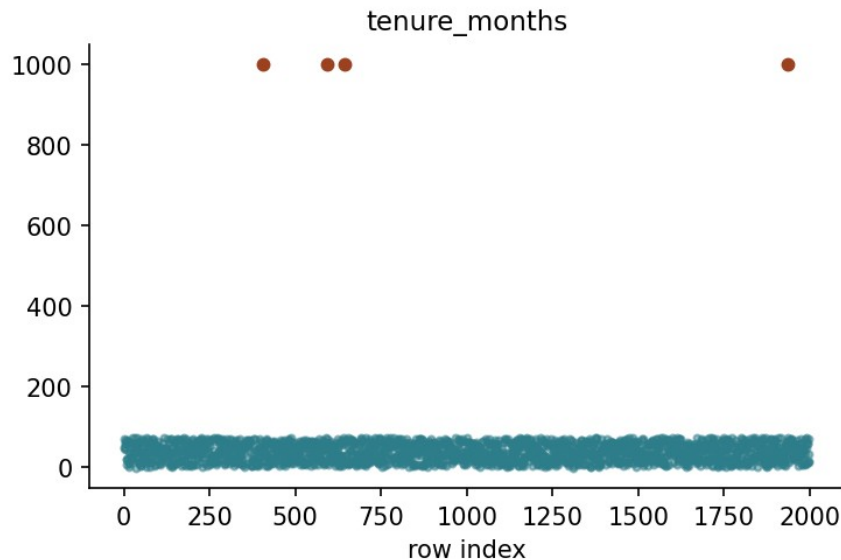
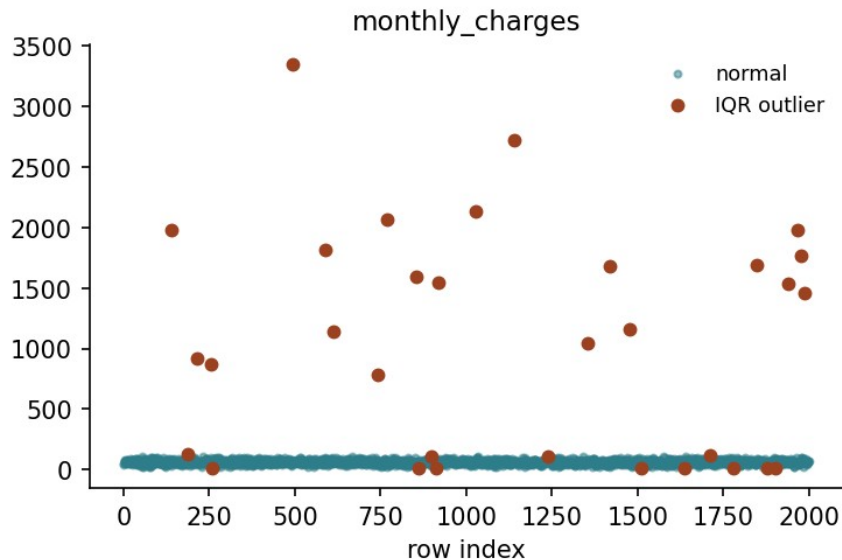
$$[Q_1 - 1.5 \text{ IQR}, \quad Q_3 + 1.5 \text{ IQR}]$$

Same neighbourhood, not the same answer



The robust rule is the one that got it wrong

What the IQR rule catches



What neither rule can tell you

`tenure_months` of **-3** is impossible. Neither rule flags it.

- -3 is not *extreme* relative to a column spanning 0-72
- It is not *unusual*. It is *invalid*: a different question entirely

Once something is flagged

Detection finds candidates. What to do next is a **domain** decision.

- **Cap** it at the fence: 3,344.7 becomes 110.0 (*Winsorising*)
- **Remove** the row
- **Correct** it, if the true value is recoverable
- **Leave** it: unusual is not the same as wrong

Notebook 1, live

section, and its slides	you call (pandas, no model yet)
first look, 6	<code>df.info()</code> , <code>df.describe()</code>
distributions, 7	<code>df.hist()</code>
missingness, 8-12	<code>df.isna().mean()</code> , and the mean before and after filling
outliers from scratch, 15-19	<code>.mean()</code> and <code>.std()</code> for rule 1, <code>.quantile([0.25, 0.75])</code> for rule 2
correlations, categories	<code>df.corr()</code> , <code>.value_counts()</code>

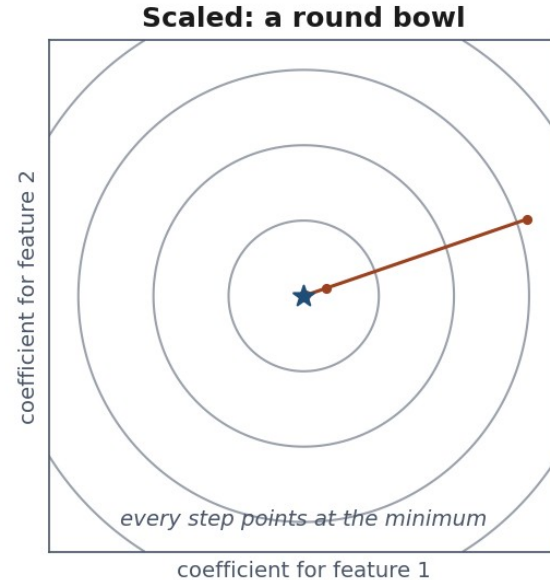
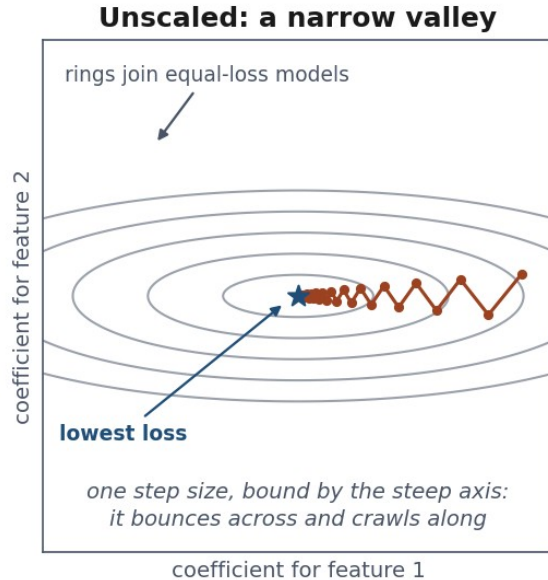
Break

Why scale?

- `tenure_months: 0-72. monthly_charges: 15-128`
- **Training** = choosing the model's numbers to make Lesson 1's average loss small
- **Gradient descent** does it by repeated small steps downhill on that loss
- One step size - the **learning rate** - along every axis, every iteration. Can it suit both columns?

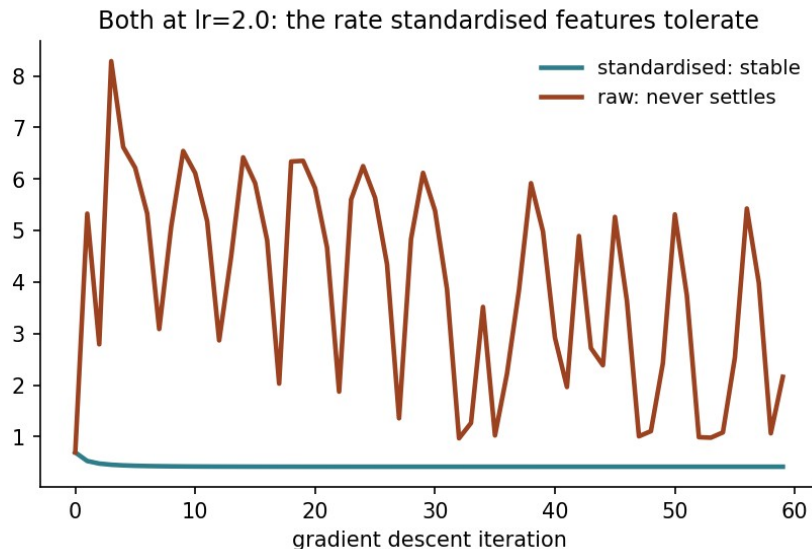
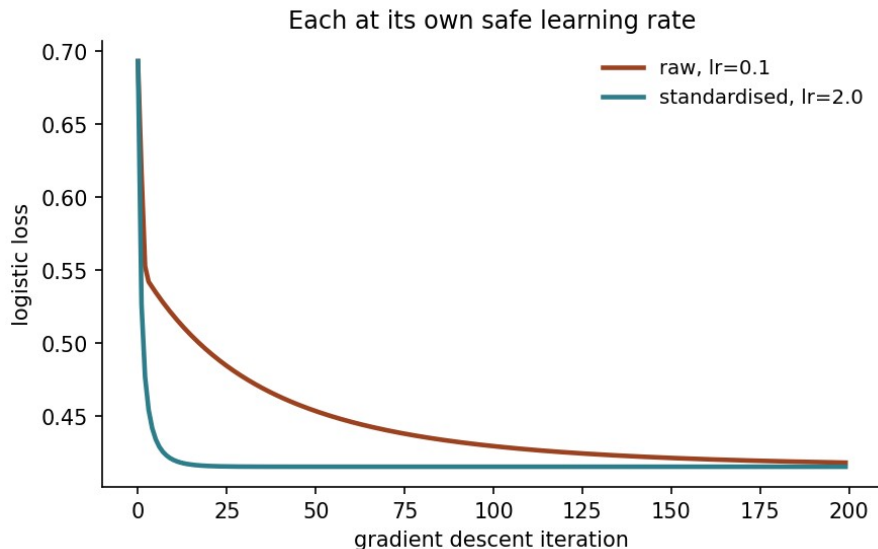
A ravine, not a bowl

One stride length, two very different directions



A 100:1 variance ratio, and what it costs

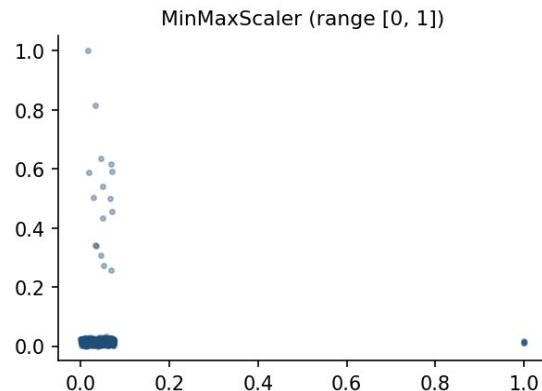
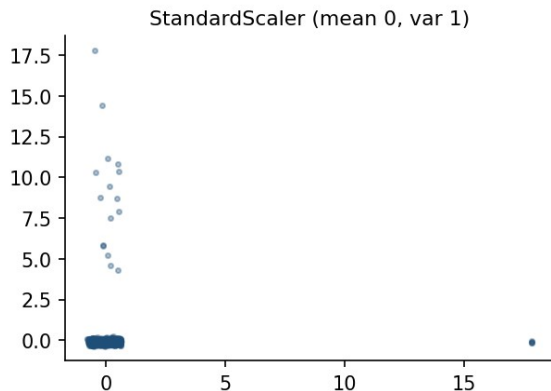
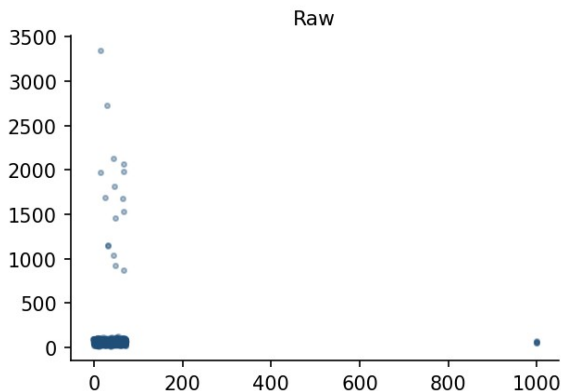
A 100:1 variance ratio, and what it costs



StandardScaler vs MinMaxScaler

$(x - \mu)/\sigma$ and $(x - \min)/(\max - \min)$, both fitted on training data only.

tenure_months (x) against monthly_charges (y)



Every encoding makes a claim about the categories

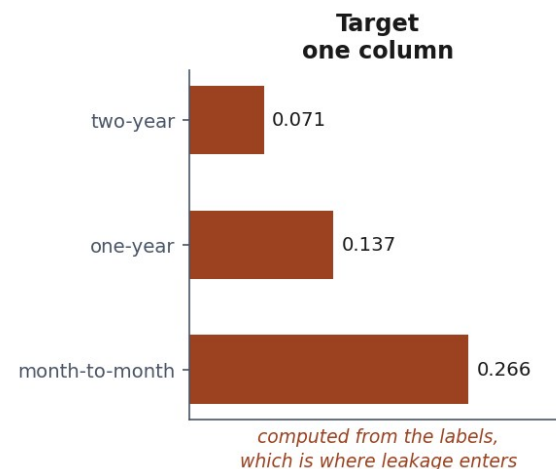
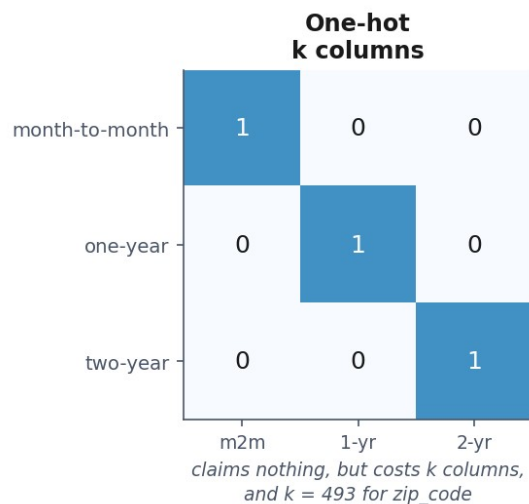
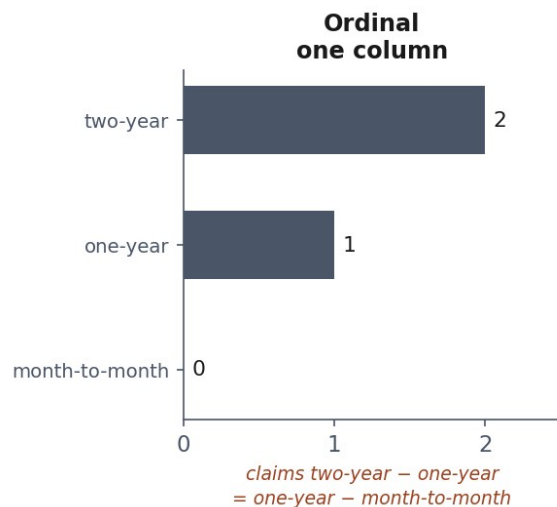
A model does arithmetic, and "month-to-month" offers none.

contract_type	rows	churn rate
month-to-month	1,092	0.266
one-year	488	0.137
two-year	420	0.071

One customer, three encodings

Encoding	A one-year customer becomes	What that asserts
One-hot	[0, 1, 0]	simply different: no order, no distances
Ordinal	1	ordered, and equally spaced
Target	0.137	the level's own average outcome stands in for it

Equal steps, unequal meaning



One-hot claims nothing, and pays in columns

- A column with k levels becomes k binary columns, one 1 per row: `OneHotEncoder`
- It asserts no order and no distance - which is why it is the default
- The price is width: `zip_code` would add **493** columns to 2,000 rows
- Target encoding compresses any cardinality into one column - computed from the labels, which is where leakage gets in

All k dummies plus an intercept cannot both be free

Most models carry a column of ones - the **intercept** - so they can fit a baseline level.

$$\sum_{j=1}^k \text{dummy}_j = 1 = \text{intercept}$$

Rank deficient by exactly one

Why all k dummies plus an intercept cannot work

contract_type	m2m	1y	2y	sum	intercept
month-to-month	1	0	0	= 1	1
one-year	0	1	0	= 1	1
two-year	0	0	1	= 1	1

the k dummy columns always sum to the intercept column: rank deficient by exactly one

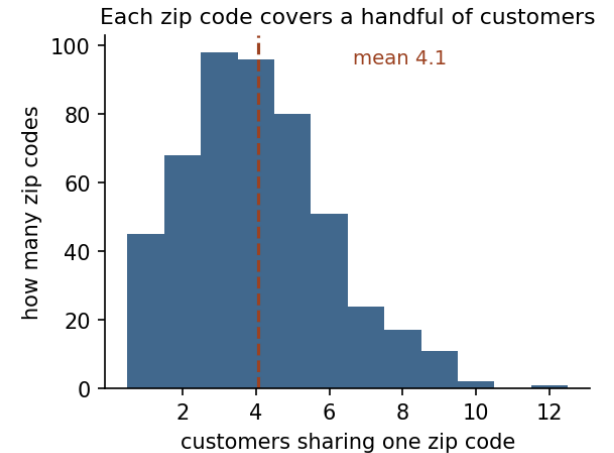
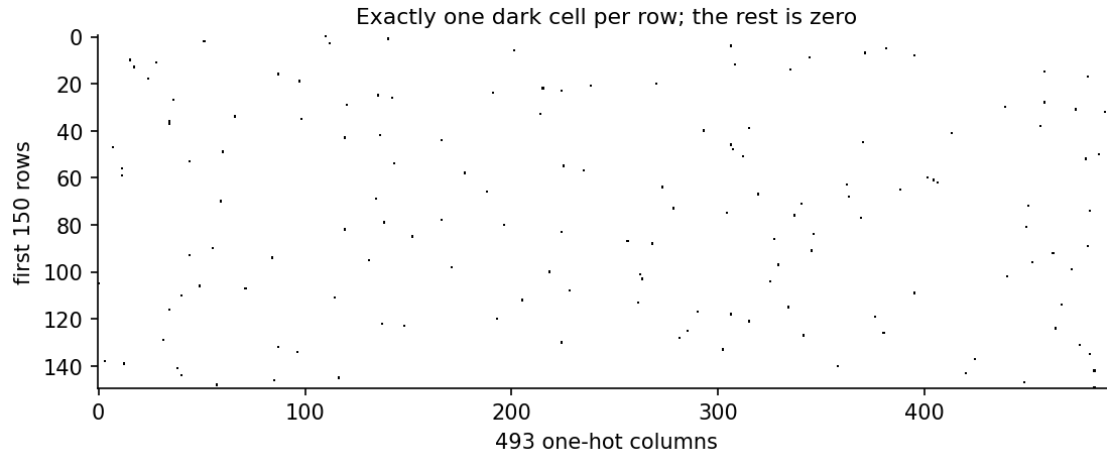
The curse of dimensionality, briefly

One-hot `zip_code`: **493** new columns, mostly zero.

- Distance-based methods (Lesson 6): everything looks equidistant
- More parameters to estimate, same sample size
→ more variance

zip_code: 493 codes, 2,000 customers

zip_code, one-hot encoded: 493 columns for 2000 rows



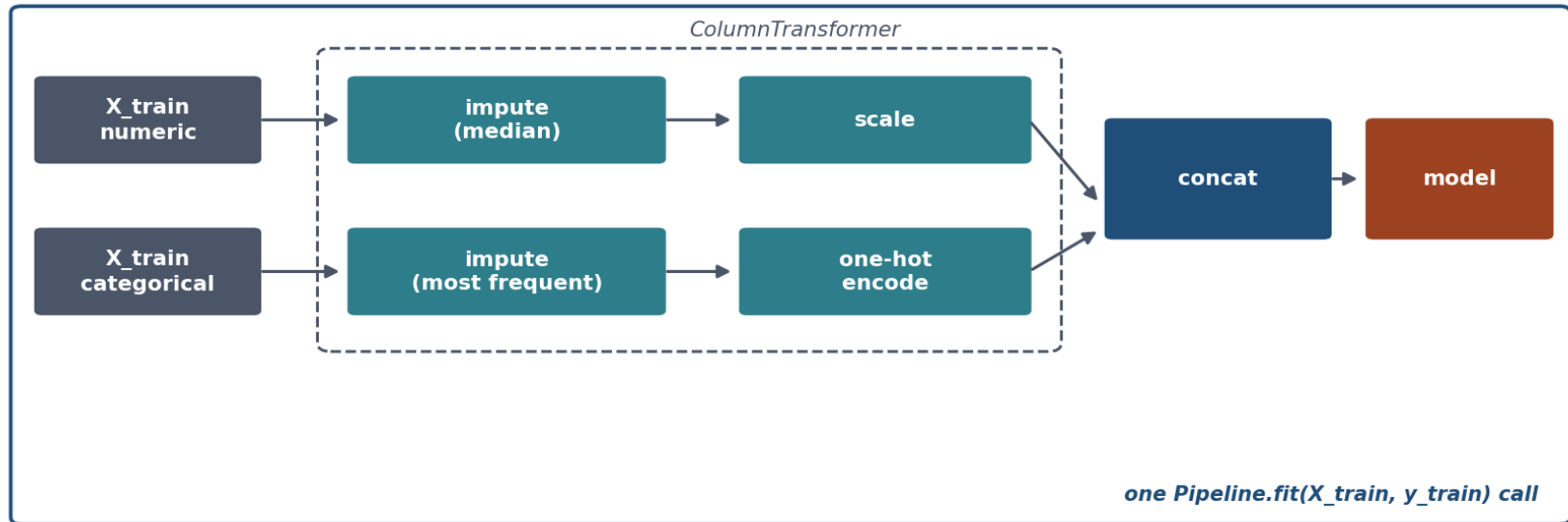
Restating Lesson 1's argument

f includes **every** learned parameter, not just “the model part”.

$$\mathbb{E}_{T \sim \mathcal{D}^m} [\hat{R}_T(f)] = R(f)$$

ColumnTransformer + Pipeline

Every learned parameter comes from inside this box



The code

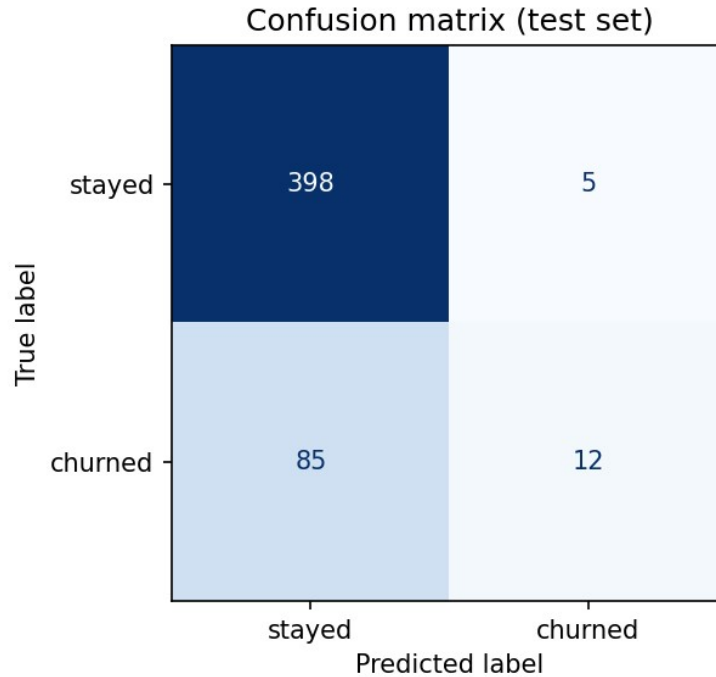
```
prep = ColumnTransformer([  
    ("num", num_pipe, num_cols),  
    ("cat", cat_pipe, cat_cols),  
])  
model = Pipeline([  
    ("prep", prep),  
    ("clf", clf),  
])  
model.fit(X_train, y_train)
```

The model, on churn

- Accuracy: baseline **0.806**, model **0.820** (numeric + low-cardinality categoricals, no zip)
- Area under the receiver operating characteristic curve (AUC): **0.751**, the chance a random churning scores above a random non-churner; 0.5 is a coin flip
- In code: `LogisticRegression` against `DummyClassifier`, `roc_auc_score`

Modest accuracy gain. Why?

Where the errors fall



Feature engineering

Ratios, interactions, binning: a linear model cannot invent these itself.

`total_paid = charge × months stayed`. AUC:
`0.7514 → 0.7548`, under four thousandths. Clean
`tenure_months` first and the same feature scores
0.7439 - the gain was living in the 999s.

In code: `FunctionTransformer(add_total_paid)`,
a step inside the pipeline

Notebook 2, live

section, and its slides	you call → the number to come back with
why scale, 24-27	gradient descent by hand → ratio 110:1 (built as 100:1); rate 2.0 scaled, 0.1 raw; raw at 2.0 swings 0.69 to 8.29
encoding, 28-33	pd.get_dummies, np.linalg.matrix_rank → 4 columns, rank 3
the pipeline, 36-38	train_test_split, then ColumnTransformer + Pipeline: SimpleImputer, StandardScaler, OneHotEncoder, LogisticRegression
the result, 39-40	DummyClassifier, accuracy_score, roc_auc_score, ConfusionMatrixDisplay
feature engineering, 41	FunctionTransformer → 0.7514 → 0.7548 → 0.7439

Same rule, broken three ways

Same rule, broken three ways: only the first one looks like a bug

Lesson 1

select features on the full dataset

visible: an obvious extra step

Today, leak 1

impute a missing value on the full dataset

invisible: looks like ordinary cleaning

Today, leak 2

encode a category on the full dataset

invisible: two unremarkable lines of code

Leak 1: impute before splitting

`KNNImputer` fills a gap with the average of the five most similar rows: its **donors**.

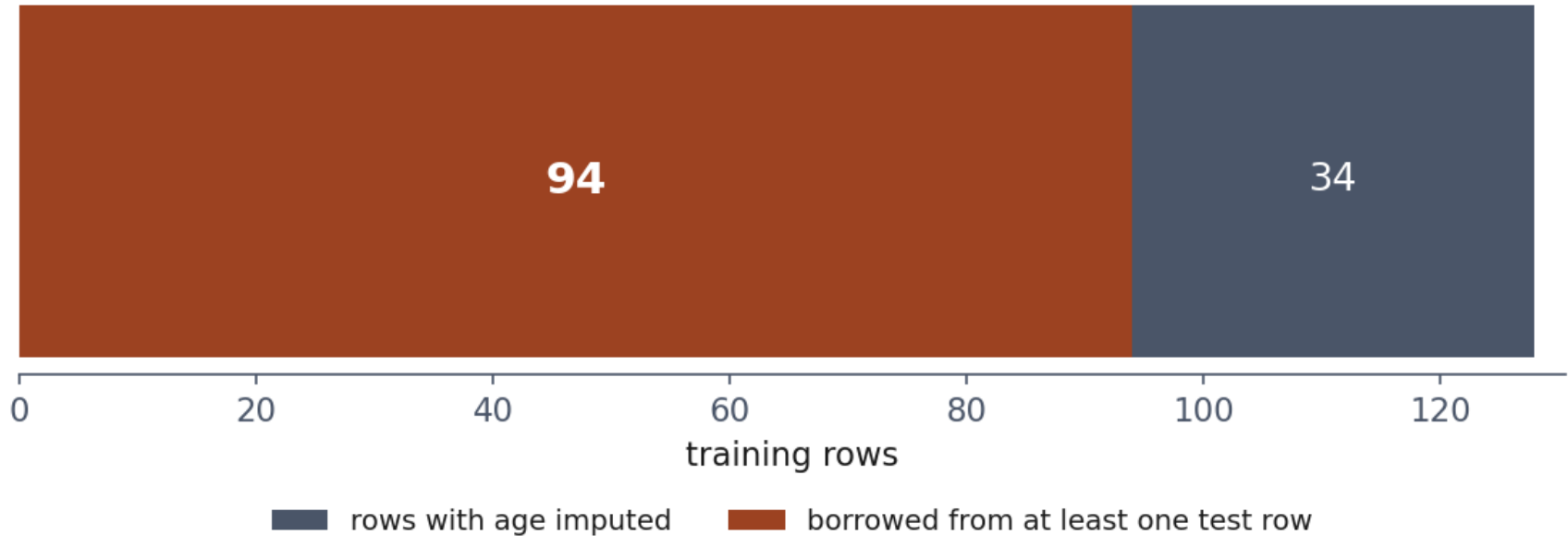
Fit it on train + test, and some of those rows are in the test set.

The smoking gun

94 of 128 training rows with missing age had a **test-set row** among the five donors used to fill it.
In code: `nan_euclidean_distances`, the imputer's own metric, names the donors.
Nothing raised an error.

Leakage, counted

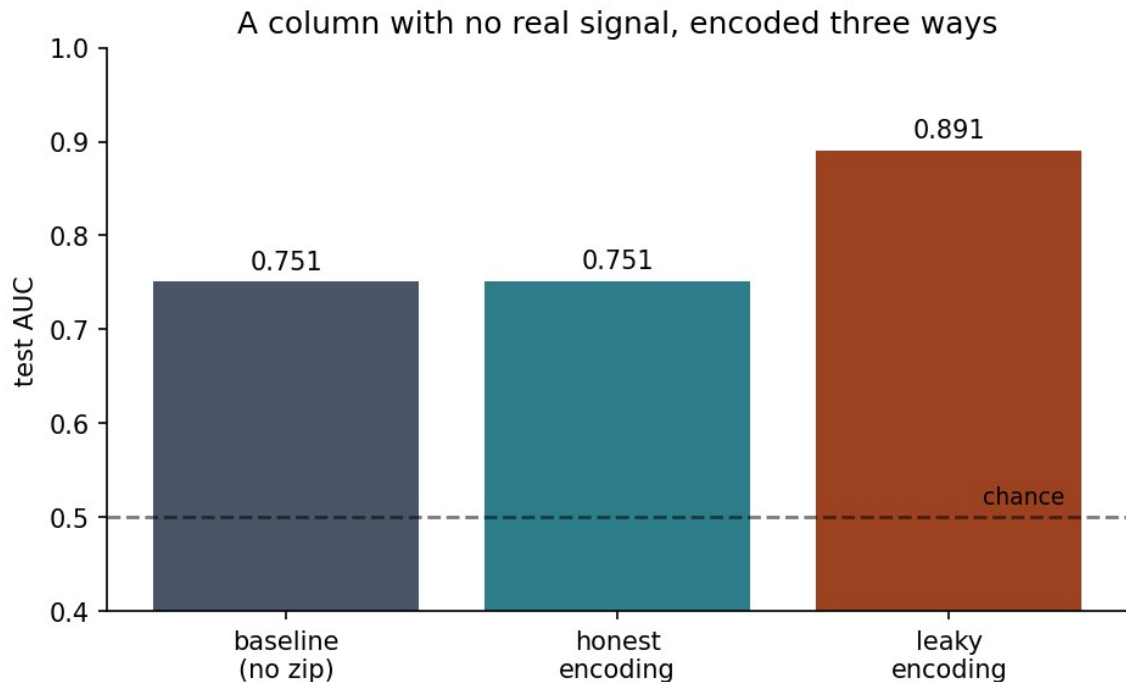
94 of 128 imputed values touched the test set



Leak 2: encode before splitting

Replace `zip_code` with the **average churn rate of its own customers**, computed before anybody has split anything.

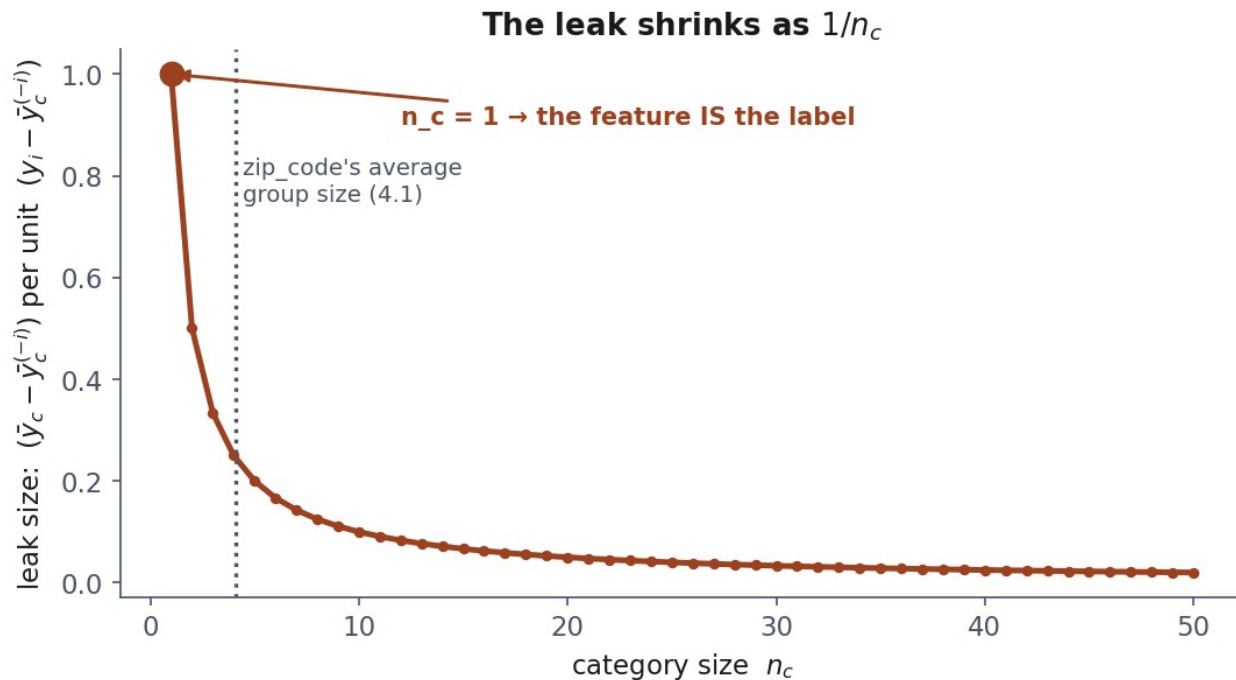
A column with no real signal, encoded three ways



Your own disagreement, divided by
your group size

$$\bar{y}_c - \bar{y}_c^{(-i)} = \frac{y_i - \bar{y}_c}{n_c}$$

The leak shrinks as $1/n_c$



The fix

`sklearn.preprocessing.TargetEncoder`, inside the pipeline.

It splits the training rows into five parts and encodes each row from the **other four**: never from its own label.

The rule, restated

Everything that **learns from data** is fitted on the training rows only, inside the pipeline.

A mean. A median. A set of neighbours. A per-category average. A set of bin edges.

Notebook 3, live

leak, and its slides	you call → what it shows
imputation, 44-46	KNNImputer fitted on every row; nan_euclidean_distances, its own metric, names the donors → 94 of 128
encoding, 47-50	churn rate per zip_code, before train_test_split → AUC 0.891 from a column with no signal
the fix, 51	TargetEncoder inside the ColumnTransformer → the honest 0.751
your own split	clone both models, refit on a split of your own
scored with	roc_auc_score throughout

What we did today

- Explored before transforming: types, gaps, shape, correlations
- Diagnosed *why* values are missing, not just *how many*
- Two outlier rules, and what neither can tell you
- Scaling, encoding, engineering: all inside Pipeline
- Two leaks that look nothing like leaking

Homework: we discuss it on Friday 16 October

Exercises/

02_data_exploration_and_preparation.md

Build the full preparation pipeline yourself, and **find a leak on purpose** before fixing it.

Before next week

- Work the three notebooks in order
- Read the handout: the derivations are examinable
- Take the quiz
- **Do the exercise**

Next: regression, the first model we derive completely.